

AI Smart Traffic Intelligence Platform

Turning any camera feed into fair, evidence-based enforcement

Startup Innovation Competition 2026 | Final

The problem

Sri Lanka records roughly **3,000 road deaths a year**, most from preventable violations.

Officers cannot watch every junction, 24 hours a day.

Violations without photographic proof or a verified plate are disputed.

Coverage and proof are the enforcement gap.

What we built

One AI pipeline detects **nine violation types** on an ordinary camera.

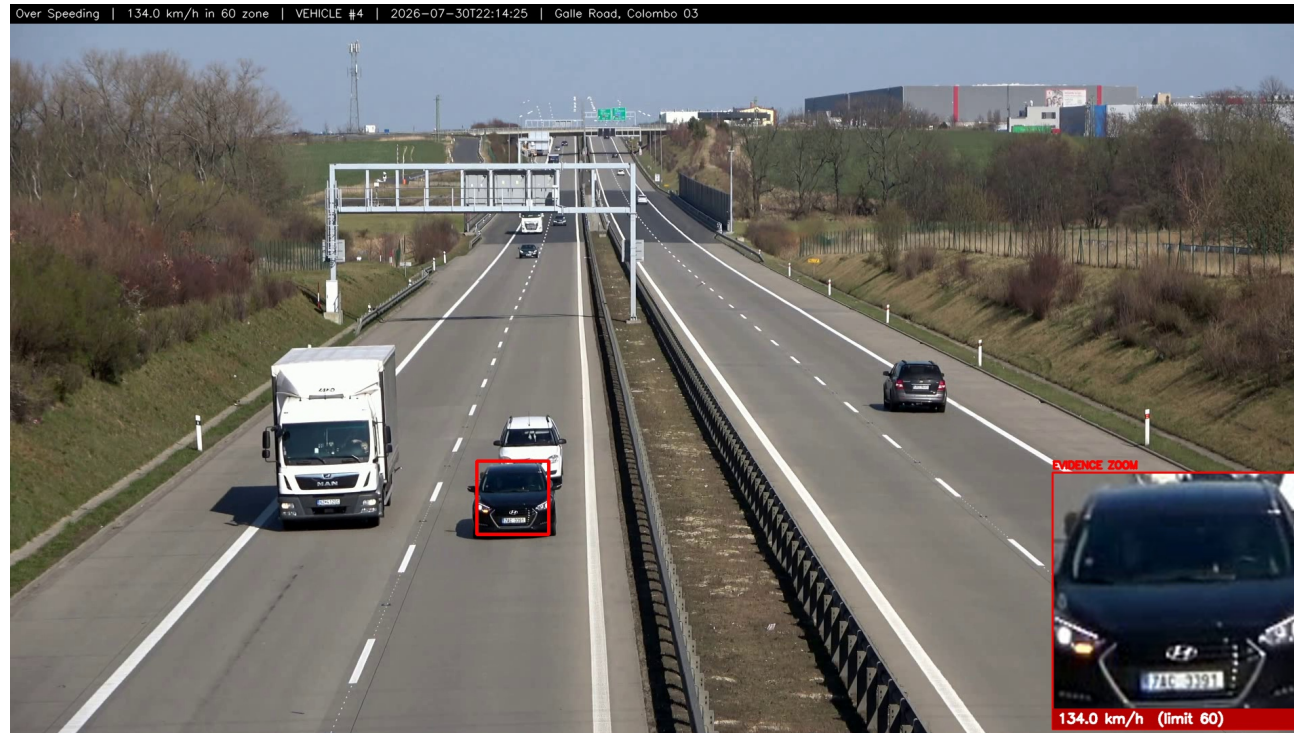
Reads **number plates on every vehicle**, not only offenders.

Measures **real speed** through geometric calibration.

Issues an **e-challan with photographic evidence** to the police.

Proof: over-speeding

Calibrated speed verified at **83-163 km/h** across two cameras.



134.0 km/h in a 60 zone - vehicle boxed, plate zoomed, proof stamped on the image.

Proof: helmet and triple riding

Both detected from a single uploaded photograph, two challans.



Three riders and no helmets on a CC-licensed public image.

Proof: plate recognition

A plate is confirmed only when **three separate reads** agree.



18 vehicles tracked; plate AAG 4002 read and confirmed.

Why it can be trusted

Parked vehicles are never fined - 41 vehicles, 0 violations.

Unreadable plates say UNREADABLE - never invented.

No calibration means no speed, not a guess.

85 automated tests, including every refusal case.

Six real false-positive defects found and fixed by testing.

Deployment

Works with **existing CCTV**, a phone stream, or a drone.

Pilot on one laptop; cities use GPU edge boxes.

Helmet, seatbelt and plate models are **drop-in files**.

Python, FastAPI, YOLOv8, ByteTrack, OpenCV, EasyOCR, SQLite.

Impact and roadmap

24/7 coverage without multiplying officers.

Court-ready evidence means fines that hold up.

Searchable plate log enables **repeat-offender analytics**.

Next: night models, vision-language for ambiguous cases, junction pilot.

Safer roads, proven by evidence.

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